Partial Observability

Tutorial 14

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13.1 *Exam Question: Partially Observable MDPs

1. In POMDPs, we usually maintain an internal state s based on the interaction history. Two ways to do that include maintaining a belief state or using frame stacking. Give an advantage and a disadvantage of belief states over frame stacking. (In case it is relevant, assume the number of possible latent system states x is much larger than the dimensionality of observations times the number of stacked frames.)

1. Advantages: Belief states are Markov, so we can use them to compute optimal policies, belief states are interpretable, belief states are reasonably compact (s has as many dimensions as x has states), belief state can be computed recursively without memorizing the history.
   Disadvantages: Belief updates are harder to implement than frame stacking, only for discrete states, underlying models are needed, underlying models can be difficult to learn, belief updates are computationally expensive.

2. While in POMDPs, we have separate latent system states x, observations o and internal agent states s, in MDPs we have $s = x = o$. In general, frame stacking does not result in an internal state that is a Markov function of a history. Is this also the case when the environment is an MDP? Explain why, using the criterion for internal Markov states in your answer.

2. In MDPs, the environment state (x) is directly observed ($x = o$). So only the last frame is already a Markov state. If we stack a couple of frames, that only adds information but s is still Markov. So: it is not the case that frame stacking result in a non-Markov internal state.

   From the definition of Markov state we can see that any time the internal state is encountered, the probability of future state is the same. Since the stack includes x, and we already know that x contains enough all information about future system behavior (x is a Markov state), the stack is so as well.

Good
luck!

Thank
You!